

Interval capacity and second-order bounds for distinct-difference configurations

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Abstract

We apply a signed interval-capacity certificate with boundary constant $2/3$ to several distinct-difference problems. A sonar sequence with n rows and m columns satisfies $m \leq n + 3(\pi^2/36)^{1/3}n^{2/3} + 4n^{1/3}$ for $n \geq 160^3$. The argument preserves the exact column marginal in Cauchy–Schwarz. We also give explicit bounds for weak Sidon sets, bounded difference multiplicity, difference triangle sets, Manhattan configurations and Sidon sets in integer boxes. All remainders and onsets are specified. For the horizontal kernel functional $f(0) \int |t|f(t) dt$, the optimum among autocorrelations of nonnegative factors is $\pi^2/32$. An explicit nonnegative positive-definite perturbation has a smaller value, so that constant is not the infimum in the full admissible class; the latter infimum remains open. Only the earlier ordinary Sidon bound, not the transfers proved here, has been formalised in Lean.

Keywords: sonar sequences; Sidon sets; distinct differences; interval capacity; signed measures; explicit bounds.

1 Results and source-qualified comparisons

We write $[N] = \{0, \dots, N-1\}$; translation gives the same results for $\{1, \dots, N\}$. A strong Sidon set in $\mathbb{Z}^d$ has distinct unordered pair sums, including repeated elements. Equivalently, every nonzero ordered difference has at most one representation: from $a - b = c - d \neq 0$ apply sum uniqueness to $a + d = c + b$, and conversely rearrange an equality of sums. All roots are positive real roots. Integer parameters need not be perfect powers. The bounds below are uniform over the stated sets.

A sonar sequence with n rows and m columns consists of $y_0, \dots, y_{m-1} \in [n]$, one point (i, y_i) in each column, with all nonzero ordered displacement vectors distinct. Repeated row values are permitted; the differences are ordinary integer differences.

Theorem 1.1 (Sonar). *For every such sequence,*

$$m \leq n + 2n^{2/3} + 3n^{1/3} \quad (n \geq 48^3), \quad (1.1)$$

$$m \leq n + 3(\pi^2/36)^{1/3}n^{2/3} + 4n^{1/3} \quad (n \geq 160^3). \quad (1.2)$$

The best proved explicit published sonar bound located in the bounded source audit is $m \leq n + 3.78n^{2/3} + 4.76n^{1/3} + 2$, for all n , in Osorio–Ruiz–Trujillo–Urbano [ORTU14, Theorem 2.1]. Erdős–Graham–Ruzsa–Taylor [EGRT92, Theorem 4] state the coefficient 5; their printed proof gives 4 with lower-order terms, and their coefficient 3 occurs in a remark without a proof. Our comparison concerns the second-order coefficient and does not assert pointwise superiority at every finite n .¹

A weak Sidon set requires uniqueness only for unordered pairs of *distinct* elements; diagonal sums are excluded from its definition.

Theorem 1.2 (Weak Sidon). *For every integer $N \geq 90^4$ and every weak Sidon $A \subseteq [N]$,*

$$|A| < \sqrt{N} + \sqrt{8/3} N^{1/4} + 2. \quad (1.3)$$

¹The main pages 104–108 of Chen–Kløve [CK96], the sonar part of Robinson [Rob85], and the Caicedo thesis were not available in the source audit. The accessible Chen–Kløve pages identify sonar as $(h, p) = (2, 1)$ and say their result generalises the EGRT bound; its exact specialisation was not determined. These access gaps preclude an exhaustive record or priority claim. Some sources interchange the letters for rows and columns; our convention is fixed above.

The best proved coefficient located in that audit is $\sqrt{3} - \eta$, with $\eta \geq 0.0089$, in a bound with $O(1)$ remainder, from Balogh–Füredi–Roy [BFR21, Theorem 5.1]. This is a citation to arXiv version 2, not an assertion that the Monthly version contains that theorem.

Theorem 1.3 (Bounded difference multiplicity). *Let $g, N \geq 1$ be integers, $gN \geq 120^4$, and $A \subseteq [N]$. If $\#\{(a, b) \in A^2 : a - b = d\} \leq g$ for every nonzero integer d , then*

$$|A| < \sqrt{gN} + \frac{2\sqrt{2}}{3}(gN)^{1/4} + 1. \quad (1.4)$$

The same statement holds with N replaced by the diameter D of A when $gD \geq 120^4$.

This is the difference-multiplicity convention, not $B_2[g]$ sum multiplicity. Balogh–Füredi–Roy [BFR21, Section 6, (6.1)] prove the coefficient 1 with additive $1/2$. Carter–Hunter–O’Bryant [CHO25] give a smaller, explicitly g -dependent coefficient $1 - \varepsilon_g/2$; their journal abstract provides $\varepsilon_g \geq 0.0062g^{-4}$. The arXiv and journal bounds are kept distinct in this comparison. Theorem 1.3 is a routine rescaling of the earlier scalar Sidon inequality.

An (n, k) difference triangle set consists of n rows of $k + 1$ increasing integer marks, each row starting at zero, with all positive within-row differences globally distinct. Its scope is the largest mark, and $m(n, k)$ is the least possible scope. Cross-row differences are not constrained.

Theorem 1.4 (Difference triangles). *For all integers $n \geq 1$, $k \geq 20365$, put $\Phi(k) = (\sqrt{4k + 8/9} - 2\sqrt{2}/3)/2$. Then*

$$m(n, k) > n\Phi(k)^4, \quad (1.5)$$

$$m(n, k) \geq n \left(k^2 - \frac{4\sqrt{2}}{3}k^{3/2} + \frac{16}{9}k - 2\sqrt{k} \right). \quad (1.6)$$

The located comparison is Kløve’s bound $n(k^2 - 2k\sqrt{k} + (k + \sqrt{k})/4)$, as quoted by Chee–Colbourn [CC97, Theorem 1] from Kløve’s 1988 Theorem 2. This is a secondary-source comparison: the 1988 and 1989 primary papers and Chen–Kløve’s 1991 lower-bound paper were not fully read. Theorem 1.4 is another routine consequence of the scalar inequality, summed over rows and inverted at a real parameter.

Theorem 1.5 (Manhattan configurations). *Let $A \subset \mathbb{Z}^2$ have unique nonzero ordered difference vectors, and $|p_1 - q_1| + |p_2 - q_2| \leq r$ for all $p, q \in A$. For every real $r \geq 160^3$,*

$$|A| \leq \frac{r}{\sqrt{2}} + (4/3)^{1/3}r^{2/3} + 9r^{1/3}. \quad (1.7)$$

The best bound located for this comparison is $r/\sqrt{2} + 3 \cdot 2^{-4/3}r^{2/3} + O(r^{1/3})$, in Blackburn–Etzion–Martin–Paterson [BEMP10, Theorem 9, arXiv version 2]. The theorem numbering of the journal version was not verified.

Theorem 1.6 (Integer boxes). *Let $d \geq 2, N \geq 1$ be integers, let $A \subseteq [N]^d$ be strong Sidon, and set*

$$x = N^{d/(2d+2)}, \quad c_d = \frac{d+1}{2}(8/9)^{d/(d+1)}. \quad (1.8)$$

If $x \geq \max(120, 4d)$, then

$$|A| \leq N^{d/2} + c_d N^{d^2/(2d+2)} + 2d^2 N^{d(d-1)/(2d+2)}. \quad (1.9)$$

For $d = 2$, every integer $N \geq 1$ also satisfies

$$|A| \leq N + (8/3)^{1/3}N^{2/3} + 18N^{1/3}. \quad (1.10)$$

For each fixed d the final term in (1.9) is lower order; no asymptotic uniformity in growing d is asserted beyond its explicit hypotheses. In dimension two the located coefficients are $3/2$ asymptotically, attributed to Robinson [Rob85], and 1.9 with remainder $1.6N^{1/3} + 1$ for all N , attributed to Caicedo’s 2016 thesis. Both were read only as secondary statements in Trujillo [Tru23, pp. 1038–1039]. For $d \geq 3$ the bounded audit did not establish the status of earlier explicit constants. These statements are source comparisons, not a claim of an exhaustive search.

The main additional inputs in this note are the exact column marginal and the product use of a signed capacity certificate. The scalar one-dimensional machinery comes from the ordinary Sidon note [Chen26]. None of these bounds resolves the original conjecture on the second-order term for ordinary Sidon sets.

2 A self-contained signed capacity certificate

We reproduce the analytic construction from [Chen26]. It supplies unit potential on a closed interval, with a finite signed measure of energy at most $L + 2/3 + 200e^{-\alpha L}$, $\alpha = \log(4/3)$. The ordinary Sidon consequence of that paper has been checked in Lean; the present analytic construction and its transfers are not claimed as new formalised theorems.

2.1 Kernel and signed energy

Lebesgue measure on the real line is denoted by dt . For an integrable function v and a measure μ , write

$$(v * \mu)(x) = \int v(x-t) d\mu(t)$$

whenever the integral exists. A finite signed real measure means one with finite total variation $\|\mu\|_{\text{TV}} = |\mu|(\mathbb{R})$.

Set

$$h(t) = 2(1-t)\mathbf{1}_{[0,1]}(t), \quad \tilde{h}(t) = h(-t), \quad f = h * \tilde{h}. \quad (2.1)$$

The specified value $h(0) = 2$ will be convenient in a pointwise identity.

Lemma 2.1 (kernel and Cauchy inequality). *The function f is even and*

$$f(x) = \begin{cases} \frac{4}{3} - 2|x| + \frac{2}{3}|x|^3, & |x| \leq 1, \\ 0, & |x| > 1. \end{cases} \quad (2.2)$$

It is nonnegative, nonincreasing on $[0, \infty)$, and satisfies

$$f(0) = \|h\|_2^2 = \frac{4}{3}, \quad \int_{\mathbb{R}} h(t) dt = 1, \quad \int_{\mathbb{R}} f(t) dt = 1. \quad (2.3)$$

For finite signed real measures define

$$E(\mu, \nu) = \iint f(x-y) d\mu(x) d\nu(y).$$

Then

$$E(\mu, \nu) = \int_{\mathbb{R}} (h * \mu)(t)(h * \nu)(t) dt, \quad E(\mu, \mu) \geq 0, \quad (2.4)$$

and

$$|E(\mu, \nu)|^2 \leq E(\mu, \mu)E(\nu, \nu). \quad (2.5)$$

Proof. The autocorrelation is even by a change of variable. For $0 \leq x \leq 1$, its value is

$$4 \int_0^{1-x} (1-s-x)(1-s) ds = \frac{4}{3} - 2x + \frac{2}{3}x^3;$$

for $x > 1$ the two supports have no overlap of positive length. This proves (2.2). Nonnegativity also follows directly from the integral of the two nonnegative factors. On $[0, 1]$ the derivative is $-2 + 2x^2 \leq 0$, and outside this interval the function is zero. Integrating h and h^2 gives (2.3); the mass of f is $(\int h)^2 = 1$ by Tonelli's theorem.

For any finite signed measure, Cauchy–Schwarz with respect to $|\mu|$ gives

$$\int |h * \mu|^2 \leq \|\mu\|_{\text{TV}} \int \int |h(t-x)|^2 d|\mu|(x) dt = \|\mu\|_{\text{TV}}^2 \|h\|_2^2.$$

Thus $h * \mu \in L^2$. Moreover, $\int |h(t-x)h(t-y)| dt \leq \|h\|_2^2$, so the corresponding triple integral against $|\mu|$ and $|\nu|$ is finite. Fubini's theorem is therefore applicable. Its inner integral is $f(x-y)$, which proves (2.4). Applying the usual Cauchy–Schwarz inequality in real L^2 proves (2.5). This argument allows either measure to be signed. $\square$

2.2 The half-line correction

Let U be the probability measure with density $v_1 = \mathbf{1}_{[0,1]}$, and let $U^{*0} = \delta_0$. Define

$$g_+ = \frac{1}{2} \sum_{n=0}^{\infty} U^{*n}. \quad (2.6)$$

The following construction proves everything needed about this measure; no renewal theorem is used.

Lemma 2.2 (local finiteness and exact potential). *The measure g_+ is locally finite, supported on $[0, \infty)$, and*

$$h * g_+ = \mathbf{1}_{[0, \infty)}, \quad (f * g_+)(x) = 1 \quad (x \geq 0). \quad (2.7)$$

Proof. For $n \geq 1$, let v_n be the density of U^{*n} . Convolution gives representatives satisfying, for $t \geq 0$,

$$0 \leq v_n(t) \leq \frac{t^{n-1}}{(n-1)!}. \quad (2.8)$$

For $n = 1$ this is the bound $v_1 \leq 1$. If it holds at n , then

$$v_{n+1}(t) = \int_0^{\min(1,t)} v_n(t-s) ds \leq \int_0^t \frac{(t-s)^{n-1}}{(n-1)!} ds = \frac{t^n}{n!}.$$

For every fixed $R \geq 0$, it follows that $U^{*n}([0, R]) \leq R^n/n!$. The sum in (2.6) consequently has finite mass on every bounded interval.

For a direct proof of the first identity in (2.7), put $H = \mathbf{1}_{[0, \infty)}$. The definitions give the pointwise identity $h/2 = H - U * H$. Thus each finite partial sum telescopes:

$$\frac{1}{2} \sum_{n=0}^m h * U^{*n} = H - U^{*(m+1)} * H. \quad (2.9)$$

For $t \geq 0$, the last term on the right is between zero and $t^{m+1}/(m+1)!$, by (2.8), and this tends to zero. For $t < 0$, both sides vanish. The nonnegative partial sums on the left converge to $h * g_+$, proving the identity. This also fixes its value at $t = 0$.

Tonelli's theorem allows association of the nonnegative convolutions. For $x \geq 0$,

$$(f * g_+)(x) = (\tilde{h} * (h * g_+))(x) = \int_{-1}^0 \tilde{h}(y) H(x-y) dy = \int_{-1}^0 \tilde{h}(y) dy = 1.$$

All the integrals against g_+ here are locally finite because their kernels have compact support. □

Write

$$g_+ = \frac{1}{2} \delta_0 + \frac{1}{2} u(t) \mathbf{1}_{[0, \infty)}(t) dt, \quad u(t) = \sum_{n=1}^{\infty} v_n(t) \quad (t \geq 0). \quad (2.10)$$

Use the right-continuous representative of u . In particular,

$$u(0) = 1, \quad u(t) = e^t \quad (0 \leq t < 1), \quad u(1) = e - 1. \quad (2.11)$$

The left limit at 1 is e , and must not be substituted for $u(1)$ in the recurrence below. To justify these regularity claims, the series in (2.10) converges uniformly on compact intervals by (2.8). For $n \geq 2$, the density $v_n(t) = \int_{t-1}^t v_{n-1}(s) ds$, with densities set to zero for $s < 0$, is continuous. Only v_1 has a jump at 1. For $0 \leq t < 1$, the induction proving (2.8) gives equality. At $t = 1$, $v_1(1) = 0$ and $v_n(1) = 1/(n-1)!$ for $n \geq 2$. These observations prove (2.11) and the stated representative convention.

Lemma 2.3 (quantitative convergence of the density). *For almost every $t \geq 0$,*

$$|u(t) - 2| \leq (e-1)(3/4)^{\lfloor t \rfloor} < 2(3/4)^{\lfloor t \rfloor}. \quad (2.12)$$

Proof. Summing the convolution recurrence for the densities yields $u = v_1 + U * u$. Thus

$$u(t) = \int_{t-1}^t u(s) ds \quad (t \geq 1). \quad (2.13)$$

This includes $t = 1$ with the right-continuous value in (2.11). On $(1, \infty)$, this identity makes u locally absolutely continuous, and $u'(t) = u(t) - u(t-1)$ almost everywhere. Fix an integer $n \geq 1$. Solving this first-order equation on $[n, n+1]$ gives

$$u(n+x) = e^x u(n) - \int_0^x e^{x-y} u(n-1+y) dy \quad (0 \leq x \leq 1).$$

Since $u(n) = \int_0^1 u(n-1+y) dy$, this becomes

$$u(n+x) = \int_0^1 K_x(y) u(n-1+y) dy, \quad K_x(y) = e^x - \mathbf{1}_{\{y \leq x\}} e^{x-y}. \quad (2.14)$$

For $0 \leq x, y \leq 1$, the kernel is nonnegative, and

$$\int_0^1 K_x(y) dy = e^x - \int_0^x e^{x-y} dy = 1.$$

For $y \in [1/2, 1]$, it also satisfies $K_x(y) \geq 1/2$. Indeed, if $y > x$, its value is $e^x \geq 1$. If $y \leq x$, its value is $e^{x-y}(e^y - 1) \geq e^{1/2} - 1 > 1/2$, using the exponential series.

Let $[m_n, M_n]$ be the smallest closed interval containing the essential range of u on $[n, n+1]$. Here essential extrema ignore sets of Lebesgue measure zero. Initially $[m_0, M_0] = [1, e]$. Equation (2.14) first implies that these closed intervals are nested. For the width estimate, every $K_x(y) dy$ contains the common measure

$$w(y) dy = \frac{1}{2} \mathbf{1}_{[1/2, 1]}(y) dy$$

of mass $1/4$. Put $c_n = \int_0^1 w(y) u(n-1+y) dy$. The remaining nonnegative measure has mass $3/4$, so every value in (2.14) lies between $c_n + (3/4)m_{n-1}$ and $c_n + (3/4)M_{n-1}$. Consequently,

$$[m_n, M_n] \subseteq [m_{n-1}, M_{n-1}], \quad M_n - m_n \leq \frac{3}{4}(M_{n-1} - m_{n-1}).$$

The nested closed intervals therefore meet in a single real number c , and

$$|u(t) - c| \leq (e-1)(3/4)^{\lfloor t \rfloor} \quad (2.15)$$

almost everywhere. This also provides a uniform essential bound and essential uniform convergence to c as $t \rightarrow \infty$.

For $t > 1$, the atom in (2.10) contributes nothing to $h * g_+(t)$, so Lemma 2.2 gives

$$1 = \frac{1}{2} \int_0^1 h(s) u(t-s) ds.$$

Letting $t \rightarrow \infty$ in this integral, (2.15) and $\int h = 1$ imply $1 = c/2$. Thus $c = 2$. Finally, $e < 3$: in its series $e = 2 + \sum_{j \geq 2} 1/j!$, the tail is strictly smaller than $\sum_{j \geq 2} 2^{-(j-1)} = 1$. This proves (2.12). $\square$

Lemma 2.4 (size, tail, and mass of the correction). *Define the signed measure supported on $[0, \infty)$*

$$q = g_+ - \mathbf{1}_{[0, \infty)}(t) dt = \frac{1}{2} \delta_0 + r(t) \mathbf{1}_{[0, \infty)}(t) dt, \quad r(t) = \frac{1}{2} u(t) - 1. \quad (2.16)$$

It is a finite signed measure, and, with $\alpha = \log(4/3)$,

$$|r(t)| \leq (3/4)^{\lfloor t \rfloor} \quad \text{almost everywhere}, \quad \|q\|_{\text{TV}} \leq \frac{9}{2}, \quad (2.17)$$

$$|q|((L, \infty)) \leq 6e^{-\alpha L} \quad (L \geq 1), \quad (2.18)$$

and

$$q(\mathbb{R}) = \frac{1}{3}. \quad (2.19)$$

Proof. The first bound follows from Lemma 2.3. Integration on unit intervals then gives

$$\|q\|_{\text{TV}} \leq \frac{1}{2} + \sum_{n=0}^{\infty} (3/4)^n = \frac{9}{2}.$$

For $L \geq 1$, the atom at zero is outside (L, ∞) , and

$$|q|((L, \infty)) \leq \sum_{n=\lfloor L \rfloor}^{\infty} (3/4)^n = 4(3/4)^{\lfloor L \rfloor} \leq \frac{16}{3}e^{-\alpha L} \leq 6e^{-\alpha L}.$$

It remains to compute the signed mass, not just its absolute bound. For $s > 0$, Tonelli's theorem and the product rule for Laplace transforms of nonnegative convolutions give

$$\begin{aligned} \int e^{-st} dg_+(t) &= \frac{1}{2} \sum_{n=0}^{\infty} \left(\int_0^1 e^{-st} dt \right)^n \\ &= \frac{1}{2(1 - (1 - e^{-s})/s)}. \end{aligned} \quad (2.20)$$

The geometric series converges since $0 < \int_0^1 e^{-st} dt < 1$. Taylor's formula for the exponential, as $s \downarrow 0$, gives

$$\phi(s) := 1 - \frac{1 - e^{-s}}{s} = \frac{s}{2} - \frac{s^2}{6} + O(s^3).$$

This use of $O(s^3)$ concerns only a limit: for $0 < s \leq 1$, Taylor's remainder after the cubic term of e^{-s} has absolute value at most $s^4/24$, which supplies the displayed bound directly. Subtracting the Laplace transform $1/s$ of half-line Lebesgue measure gives

$$\int e^{-st} dq(t) = \frac{1}{2\phi(s)} - \frac{1}{s} = \frac{s - 2\phi(s)}{2s\phi(s)} \longrightarrow \frac{1}{3}.$$

Because q has finite total variation and is supported on the nonnegative half-line, $|e^{-st}| \leq 1$ there. Dominated convergence for the finite measure $|q|$ shows that the limit is $q(\mathbb{R})$, proving (2.19). $\square$

2.3 Joining two half-lines

Lemma 2.5 (finite-interval energy bound). *Let $L \geq 1$, and let μ be any finite positive measure supported on the closed interval $[0, L]$. If $k = \mu(\mathbb{R})$, then*

$$E(\mu, \mu) \geq \frac{k^2}{L + \frac{2}{3} + 200e^{-\alpha L}}, \quad \alpha = \log(4/3). \quad (2.21)$$

Proof. Let q_L be the pushforward of q by the reflection $t \mapsto L - t$, and set

$$\nu_L = \mathbf{1}_{[0, L]}(t) dt + q + q_L. \quad (2.22)$$

This is a finite signed real measure, with

$$\nu_L(\mathbb{R}) = L + \frac{2}{3}. \quad (2.23)$$

Let g_-^L denote the same reflection of g_+ . As locally finite measures,

$$\nu_L = g_+ + g_-^L - dt. \quad (2.24)$$

Indeed, the two half-line Lebesgue measures sum to full-line Lebesgue measure plus Lebesgue measure on $[0, L]$; endpoints do not change those Lebesgue measures. The atoms of q at 0 and of q_L at L remain included

in (2.22). Since f is compactly supported, (2.24) may be convolved locally with f . Lemma 2.2 and evenness of f give, for $0 \leq x \leq L$,

$$V_L(x) := (f * \nu_L)(x) = 1 + 1 - \int f = 1. \quad (2.25)$$

This holds also at both endpoints. For every real x , nonnegativity and mass one of f , together with Lemma 2.4, imply

$$|V_L(x)| \leq 1 + \|f\|_\infty(\|q\|_{\text{TV}} + \|q_L\|_{\text{TV}}) \leq 1 + \frac{4}{3} \cdot 9 = 13. \quad (2.26)$$

Outside $[0, L]$, the only measure in (2.22) on (L, ∞) is the tail of q ; on $(-\infty, 0)$ it is the reflection of the same tail. In particular there are no excluded endpoint atoms. Thus (2.18) gives

$$|\nu_L|(\mathbb{R} \setminus [0, L]) \leq 12e^{-\alpha L}.$$

By (2.23), (2.25), and (2.26),

$$\begin{aligned} \left| E(\nu_L, \nu_L) - \left(L + \frac{2}{3} \right) \right| &= \left| \int_{\mathbb{R} \setminus [0, L]} (V_L(x) - 1) d\nu_L(x) \right| \\ &\leq 14 \cdot 12e^{-\alpha L} = 168e^{-\alpha L} \leq 200e^{-\alpha L}. \end{aligned} \quad (2.27)$$

Also $E(\mu, \nu_L) = \int V_L d\mu = k$. Lemma 2.1 now gives

$$k^2 \leq E(\mu, \mu)E(\nu_L, \nu_L) \leq E(\mu, \mu) \left(L + \frac{2}{3} + 200e^{-\alpha L} \right).$$

The denominator is positive and $E(\mu, \mu) \geq 0$, proving (2.21). There is no assumption that ν_L is positive. $\square$

3 Sonar and the exact column marginal

Write $a_2 = 4/3$, $b = 2/3$, $\alpha = \log(4/3)$ and $C(L) = L + b + 200e^{-\alpha L}$. Let g be an even nonnegative positive-definite horizontal kernel, $\mu = \sum_i \delta_{(i, y_i)}$, $\lambda = \sum_{i=0}^{m-1} \delta_i$, and let ρ be the pushforward of $\nu_{n/U}$ by $t \mapsto Ut$, without multiplying its mass. For $G(x, y) = g(x/T)f(y/U)$ set

$$\Lambda = \sum_{i,j=0}^{m-1} g((i-j)/T), \quad E_G(\mu, \lambda \otimes \rho) = \Lambda, \quad E_G(\lambda \otimes \rho, \lambda \otimes \rho) = \Lambda E_f(\nu_{n/U}, \nu_{n/U}). \quad (3.1)$$

When $m > 0$, $g(0) > 0$ ensures $\Lambda > 0$. Signed Cauchy–Schwarz gives

$$\Lambda \leq C(n/U)E_G(\mu, \mu) \quad (0 < U \leq n). \quad (3.2)$$

For the autocorrelations below this is an L^2 Gram inequality: the scaled factors are $T^{-1/2}h_g(\cdot/T)$ and $U^{-1/2}h(\cdot/U)$. Bounded kernels and finite total variations justify the product integrals. Kernel nonnegativity, separately, allows completion of missing differences. The exact column marginal makes (3.1) possible. A prescribed marginal in another product problem gives the same argument; an unrestricted projection cannot be substituted for a prescribed one.

3.1 The triangular horizontal kernel

Take $g(t) = (1 - |t|)_+$ and an integer $1 \leq V \leq m$. Exact summation gives

$$\Lambda = mV - \frac{V^2 - 1}{3}, \quad E_G(\mu, \mu) \leq \frac{4}{3}m + (V - 1)(U + 4/3). \quad (3.3)$$

Indeed $\sum_{d \neq 0} g(d/V) = V - 1$ and $\sum_{e \in \mathbb{Z}} f(e/U) \leq U + 4/3$. Every off-diagonal sonar difference has nonzero first coordinate and occurs once. The formula for Λ is $m + 2 \sum_{d=1}^{V-1} (m - d)(1 - d/V)$.

Put $x = n^{1/3} \geq 48$, $U = x^2$ and $V = \lceil 2x^2 \rceil \leq n$. If $m < V$ the result is immediate; otherwise the preceding inequalities apply. As $\alpha \geq 1/4$, $x^2 e^{-\alpha x}$ decreases for $x \geq 8$. The exact comparisons $(3/4)^8 < 1/9$ and $200 \cdot 48^2 < 9^6$ imply $\varepsilon = 200e^{-\alpha x} < x^{-2}$. With $C = x + b + \varepsilon$, solving the sandwich linearly gives

$$m(1 - a_2 C/V) \leq V/3 - 1/(3V) + C(1 - 1/V)(x^2 + a_2). \quad (3.4)$$

Set

$$q = \frac{2}{3x} + \frac{4}{9x^2} + \frac{2}{3x^4}, \quad B = x^3 + \frac{4}{3}x^2 + \frac{5}{6}x + \frac{20}{9} + \frac{1}{4x} + \frac{4}{3x^2}.$$

Then $a_2 C/V \leq q < 1/2$, and the right side of (3.4) is at most B . For the latter use $V \leq 2x^2 + 1$, $C \leq x + b + x^{-2}$, $C \geq x$, and $x^3/(2x^2 + 1) \geq x/2 - 1/(4x)$. With $Y = x^3 + 2x^2 + 3x$,

$$36x^3\{Y(1-q) - B\} = 14x^4 - 184x^3 - 81x^2 - 96x - 72 = 14t^4 + 712t^3 + 12591t^2 + 85376t + 141496, \quad t = x - 16.$$

It is positive for $x \geq 48$. Division by $1 - q > 0$ proves (1.1), without a preliminary asymptotic bound on m .

3.2 The cosine autocorrelation

Let $h_c(t) = (\pi/2) \sin(\pi t) \mathbf{1}_{[0,1]}(t)$ and $g = h_c * \tilde{h}_c$. For $0 \leq s \leq 1$, direct integration gives

$$g(s) = \frac{\pi^2}{8}(1-s) \cos(\pi s) + \frac{\pi}{8} \sin(\pi s), \quad a = g(0) = \frac{\pi^2}{8}, \quad M = \int |s|g(s) ds = \frac{1}{4}. \quad (3.5)$$

Extend evenly and by zero outside $[-1, 1]$. The mass is one and $g'(s) = -(\pi^3/8)(1-s) \sin(\pi s) \leq 0$ on $[0, 1]$. The moment also follows from the cumulative distribution $F(s) = (1 - \cos(\pi s))/2$: $M = 2 \int F(1-F) = 1/4$.

For any even decreasing nonnegative mass-one kernel w ,

$$T - w(0) \leq \sum_{d \in \mathbb{Z}} w(d/T) \leq T + w(0), \quad \sum_{d \neq 0} w(d/T) \leq T.$$

For $p(s) = |s|g(s)$, integration on $(0, \infty)$ gives $\int |p'| \leq \int g + \int s|g'| = 1$; hence $\text{Var}(p) \leq 2$. Comparing each sample to its adjacent cell integral yields $|\sum_d p(d/T) - TM| \leq 2$. For $T \leq m$ these estimates give

$$\Lambda \geq m(T - a) - MT^2 - 2T, \quad (3.6)$$

$$E_G(\mu, \mu) \leq aa_2 m + T(U + a_2). \quad (3.7)$$

Thus the fully finite, linear inequality is

$$m[1 - a/T - aa_2 C(n/U)/T] \leq MT + 2 + C(n/U)(U + a_2). \quad (3.8)$$

Put $x = n^{1/3}$, $v = (\pi^2/36)^{1/3}$, $T = 4vx^2$ and $U = (3/2)vx^2$. The elementary bounds $3 < \pi < 22/7$ give $3/5 < v < 2/3$ and $a = (9/2)v^3$. If $m < n$ there is nothing to prove; otherwise $x \geq 160$ ensures $T \leq n \leq m$, $U \leq n$. As $\alpha = \int_1^{4/3} dt/t \geq 1/4$,

$$\varepsilon = 200e^{-\alpha n/U} \leq 200e^{-x/4} \leq x^{-4} \quad (x \geq 160). \quad (3.9)$$

Indeed $x^4 e^{-x/4}$ decreases for $x \geq 16$ and $200 \cdot 160^4 = 131072000000 < 2^{40} < e^{40}$. Set $d_0 = (17/8)v^2$. The denominator and right side in (3.8) are $1 - u$ and B_c , where

$$u = \frac{v}{x} + \frac{d_0 + (3/2)v^2\varepsilon}{x^2}, \quad B_c = x^3 + 2vx^2 + \frac{8}{9v}x + \frac{26}{9} + \varepsilon \left(\frac{3}{2}vx^2 + \frac{4}{3} \right).$$

We have $0 < u < 1/2$. For $Y = x^3 + 3vx^2 + 4x$, exact expansion gives

$$\begin{aligned} Y(1-u) - B_c &= \left(4 - \frac{41}{8}v^2 - \frac{8}{9v} \right) x - 4v - 3vd_0 - 4d_0/x - 26/9 \\ &\quad - \varepsilon \left(\frac{3}{2}v^2x + \frac{9}{2}v^3 + \frac{6v^2}{x} + \frac{3}{2}vx^2 + \frac{4}{3} \right). \end{aligned}$$

The coefficient of x is at least $13/54$, $4v + 3vd_0 + 26/9 \leq 67/9$, and $4d_0/x \leq 1$. The final error is at most $x^{-2} + (2/3)x^{-3} + (8/3)x^{-4} + (8/3)x^{-5} < 1$. Consequently $Y(1-u) - B_c \geq 13x/54 - 85/9 > 0$. This proves (1.2) for every real $x \geq 160$. Finally $3v < 2$ follows from $(22/7)^2 < 32/3$.

3.3 The first-moment functional and its precise scope

Proposition 3.1. *For $h \geq 0$ in $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ with $\int h = 1$, let $g = h * \tilde{h}$, $a = g(0)$ and $M = \int |t|g(t) dt$. Then*

$$aM \geq \frac{\pi^2}{32}. \quad (3.10)$$

Equality is attained precisely by translates and dilates of the sine density h_c above.

Proof. The case $M = \infty$ is immediate. With $F(x) = \int_{-\infty}^x h(t) dt$, Tonelli applied to thresholds separating two points gives $M = 2 \int F(1 - F)$. Cauchy–Schwarz and the absolutely continuous change of variables $u = F(x)$ give

$$aM = 2 \left(\int h^2 \right) \left(\int F(1 - F) \right) \geq 2 \left(\int h \sqrt{F(1 - F)} \right)^2 = 2 \left(\int_0^1 \sqrt{u(1 - u)} du \right)^2 = \frac{\pi^2}{32}.$$

The change of variables remains valid where h vanishes. Equality requires $h = c\sqrt{F(1 - F)}$ almost everywhere. On the interval where $0 < F < 1$, integration of this differential equation gives the sine density on one interval; outside it $h = 0$ almost everywhere. Translation and dilation are the only freedoms. $\square$

The leading expression in (3.8) is $MT + b_y U + aa_y n^2 / (TU)$. Its AM–GM minimum is $3[(aM)(a_y b_y)]^{1/3} n^{2/3}$. For the fixed vertical ramp, $a_y b_y = 8/9$, so (3.10) proves that the best leading coefficient of this particular template over nonnegative horizontal factors is $3(\pi^2/36)^{1/3}$, when its lattice errors are lower order. This does not concern non-product two-dimensional kernels or inequalities using additional information.

For completeness, the separate one-dimensional theorem of [Chen26] says that if $w \in C_0(\mathbb{R}) \cap L^1(\mathbb{R})$ is even, nonnegative, of mass one and $w(0) > 0$, and $C_w(L)$ is the reciprocal minimum energy of positive probability measures on $[0, L]$, then $\liminf_{L \rightarrow \infty} (C_w(L) - L) \geq 8/(9w(0))$. Thus a finite intercept $C_w(L) = L + b_w + o(1)$ satisfies $w(0)b_w \geq 8/9$. Varying the vertical kernel preserves the preceding template conclusion only under these hypotheses and the sampling and certificate asymptotics needed to justify the template.

Proposition 3.2. *Every continuous, even, nonnegative positive-definite integrable kernel g with mass one satisfies*

$$g(0) \int |t|g(t) dt \geq \frac{1}{4} + \frac{1}{18\pi^2}. \quad (3.11)$$

Proof. Put $a = g(0) > 0$ and $q(t) = g(t/a)/a$. Then $q(0) = \int q = 1$ and $\int |t|q = a \int |t|g$. Positive definiteness gives $0 \leq q \leq 1$ and $\hat{q} \geq 0$, with Fourier phase $e^{-2\pi i \xi t}$. The latter follows either from the positive-definite quadratic form and truncated exponentials, or from its Fourier representation. For $B = \mathbf{1}_{[-1/2, 1/2]}$, $\hat{B}(3/2) = -2/(3\pi)$, whence $\|q - B\|_1 \geq 2/(3\pi)$. Let $\eta = \int_{|t| > 1/2} q = \int_{|t| \leq 1/2} (1 - q) \geq 1/(3\pi)$. Then

$$\int |t|q - \frac{1}{4} = \int_{|t| > 1/2} (|t| - 1/2)q + \int_{|t| \leq 1/2} (1/2 - |t|)(1 - q) \geq \frac{\eta^2}{2}.$$

Each integral is at least $\eta^2/4$: for a density at most one, the distance distribution from two boundary points has at most $2s$ mass within distance s , so integrate $(\eta - 2s)_+$. The inner strips fit because $\eta \leq 1$. $\square$

3.4 An admissible perturbation below the restricted optimum

Proposition 3.3. *The constant $\pi^2/32$ is not the infimum over the full class in Proposition 3.2. In fact, set*

$$\delta = \frac{1}{16}, \quad \theta = \frac{1}{10^7}, \quad b_\delta(t) = \delta^{-1}g(t/\delta), \quad g_\theta(t) = g(t) + \theta[b_\delta(t - 2) + b_\delta(t + 2) - 2b_\delta(t)], \quad (3.12)$$

where g is the cosine kernel (3.5). This kernel is continuous, even, nonnegative, positive definite, and of mass one, and, with $R = 1 - (129/8)\theta - 508\theta^2$,

$$g_\theta(0) \int |t|g_\theta(t) dt = \frac{\pi^2}{32}R < \frac{\pi^2}{32}. \quad (3.13)$$

Consequently the full-class infimum lies in $[1/4 + 1/(18\pi^2), (\pi^2/32)R]$; its exact value remains open.

Proof. Write $a_0 = \pi^2/8$. The two positive tails are disjoint from $[-1, 1]$. The only negative correction is in $|t| \leq \delta$, where $g(t) \geq a_0(1 - \delta) \cos(\pi\delta) \geq 465a_0/512 > a_0/2$, using $\pi\delta < 1/4$ and $\cos u \geq 1 - u^2/2$. The subtraction is at most $32\theta a_0 < a_0/2$. Thus $g_\theta \geq 0$, and its mass remains one. Direct moments give

$$a = g_\theta(0) = a_0(1 - 32\theta), \quad M = \int |t|g_\theta(t) dt = \frac{1}{4} + \frac{127}{32}\theta,$$

which proves (3.13) and $0 < R < 1$.

The Fourier transforms are

$$\widehat{h}_c(\xi) = \frac{e^{-\pi i \xi} \cos(\pi \xi)}{1 - 4\xi^2}, \quad \widehat{g}(\xi) = \frac{\cos^2(\pi \xi)}{(1 - 4\xi^2)^2}, \quad \widehat{g}_\theta(\xi) = \widehat{g}(\xi) - 4\theta \sin^2(2\pi \xi) \widehat{g}(\delta \xi),$$

with removable values at $\xi = \pm 1/2$. The distributional second derivative of the zero extension of h_c has total variation $2\pi^2$: π^2 from its endpoint derivative jumps and π^2 from the interior. Hence $|\widehat{h}_c(\xi)| \leq \min(1, 1/(2\xi^2))$. Where $\widehat{g} \neq 0$, the ratio of the subtracted term before θ to $\widehat{g}$ is

$$16 \sin^2(\pi \xi) (1 - 4\xi^2)^2 \widehat{g}(\delta \xi) \leq 16(1 + 2/\delta^2)^2 = 4210704.$$

To obtain the bound, split at $\xi^2 = 1/(2\delta^2)$ and use respectively $\widehat{g} \leq 1$ and $\widehat{g}(\delta \xi) \leq 1/(4\delta^4 \xi^4)$. Continuity handles the zeros and removable points. Since $4210704\theta < 1$, the transform is nonnegative. Its integrable fourth-power decay justifies Fourier inversion and proves positive definiteness. By Proposition 3.1, this example cannot have a nonnegative unit-mass factor in $L^1 \cap L^2$. $\square$

A secondary consequence, with a larger explicit remainder, is

$$m \leq n + 3[(\pi^2/36)R]^{1/3} n^{2/3} + 8n^{1/3}, \quad n \geq 160^3. \quad (3.14)$$

Here are the error details. For the perturbed kernel, $1 \leq a \leq 4/3$, $1/4 \leq M \leq 1/3$, its support lies in $[-33/16, 33/16]$, and

$$\text{Var}(g_\theta) \leq 2a_0 + 128\theta a_0 < 3, \quad \text{Var}(|t|g_\theta) \leq 2 + \theta(6 + 132a_0) < 3.$$

For the second estimate, each shifted $|t|b_\delta(t \mp 2)$ has variation at most $1 + (2 + \delta)2a_0/\delta$, and the central term has variation at most 2. Adjacent-cell comparisons give $|\sum_d g_\theta(d/T) - T| \leq 3$ and $\sum_d |d|g_\theta(d/T) \leq MT^2 + 3T$. The nonzero horizontal sum is at most $T + 3 - a \leq T + 2$. The same marginal argument, now justified by the nonnegative integrable Fourier transform for signed measures, yields

$$m[1 - 3/T - (4a/3)C/T] \leq MT + 3 + C(1 + 2/T)(U + 4/3), \quad C = n/U + 2/3 + \varepsilon, \quad (3.15)$$

provided $(33/16)T \leq m$, $U \leq n$. Put $x = n^{1/3}$, $v = [(\pi^2/36)R]^{1/3}$, $t_1 = v/M$, $t_2 = 3v/2$, $T = t_1 x^2$, $U = t_2 x^2$. We have $3/5 < v < 2/3$, $9/5 < t_1 < 8/3$, $9/10 < t_2 < 1$; these follow from $R > 9/10$ and the preceding rational bounds. If $m \geq n$ and $x \geq 160$, the support condition holds since $(33/16)t_1 x^2 \leq (11/2)x^2 \leq n$. The tail again obeys $\varepsilon \leq x^{-4}$. In (3.15) the denominator is $1 - u$, where

$$u = v/x + (d_1 + e_1 \varepsilon)/x^2, \quad d_1 = \frac{3 + 8a/9}{t_1} \leq \frac{565}{243} < 3, \quad e_1 = \frac{4a/3}{t_1} \leq \frac{80}{81} < 1.$$

It is positive and $u < 1/2$. The right side is at most $x^3 + 2vx^2 + (70/27)x + 8$: the x terms have coefficients $4/(3t_2) + 2/t_1 \leq 70/27$; the constants $3 + 8/9 + 2bt_2/t_1$ are at most $125/27$; and the remaining terms $\varepsilon t_2 x^2 + (4/3)\varepsilon + 2(4/3)/(t_1 t_2 x) + 2(4/3)(b + \varepsilon)/(t_1 x^2) + 2\varepsilon t_2/t_1$ are together less than 3 at $x \geq 160$. For $Y = x^3 + 3vx^2 + 8x$, substitution of $v \leq 2/3$, $d_1 \leq 3$, $e_1 \varepsilon \leq x^{-4}$ gives

$$Y(1 - u) - [x^3 + 2vx^2 + (70/27)x + 8] \geq \frac{29}{27}x - 22 > 0.$$

This proves (3.14); the case $m < n$ is immediate. The exact arithmetic bounds, rather than a fitted numerical optimum, are used throughout. The smaller coefficient in (3.14) does not imply a smaller bound at every finite n .

4 One-dimensional corollaries

For the ramp, monotonicity gives $2 \sum_{d \geq 1} f(d/T) \leq T$ for every real $T > 0$. We first record the real-parameter scalar step so that subsequent substitutions do not assume integrality.

Lemma 4.1. *Let $X \geq 120^4$, $k \geq 0$ be real, put $x = X^{1/4}$, $S = \sqrt{2}x^3$, and suppose*

$$k^2 \leq \left(X/S + 2/3 + 200e^{-\alpha X/S} \right) (S + 4k/3). \quad (4.1)$$

Then $k < x^2 + (2\sqrt{2}/3)x + 1$.

Proof. Write $\gamma = 2\sqrt{2}/3$ and $\varepsilon = 200e^{-\alpha x/\sqrt{2}}$. The scale obeys $0 < S \leq X$, $X/S \geq 1$. Since $\alpha/\sqrt{2} \geq 1/6$ and $xe^{-x/6}$ decreases for $x \geq 6$,

$$\varepsilon x \leq 24000e^{-20} < 24000/2^{20} < 1/32.$$

Expansion of (4.1) gives $P_\varepsilon(k) \leq 0$, where

$$P_\varepsilon(z) = z^2 - (\gamma x + 8/9 + 4\varepsilon/3)z - x^4 - \gamma x^3 - \sqrt{2}\varepsilon x^3.$$

At $y = x^2 + \gamma x + 1$,

$$P_0(y) = \frac{10}{9}x^2 + \frac{\gamma}{9}x + \frac{1}{9} \geq x^2, \quad \varepsilon(4y/3 + \sqrt{2}x^3) < x/8 + 3x^2/64 \leq 11x^2/64.$$

Here $y \leq 3x^2$. Thus $P_\varepsilon(y) > 0$. Its constant term is negative, so this monic quadratic has exactly one positive root, which lies between k and y . $\square$

4.1 Weak Sidon sets

For $d > 0$ let $r(d) = \#\{(x, y) \in A^2 : x - y = d\}$. If two distinct lower endpoints $p < q$ represent d , then $(p + d) + q = (q + d) + p$. Unless $q = p + d$, both pairs consist of distinct elements, are different, and contradict weak Sidonicity. Thus repeated differences are exactly adjacent edges of a three-term progression. Three lower endpoints are impossible, so $r(d) \leq 2$. Distinct repeated differences have distinct middle points: otherwise two distinct off-diagonal pairs have the same sum twice that point. These middle points are neither the minimum nor the maximum of A . Consequently, with $k = |A| \geq 2$,

$$E_f(\mu, \mu) \leq \frac{4}{3}k + T + \frac{8}{3}(k - 2) = T + 4k - 16/3 \leq T + 4k, \quad \mu = \sum_{a \in A} \delta_{a/T}. \quad (4.2)$$

For $k = 0, 1$ the last bound holds directly. The diagonal energy remains $4k/3$ even though diagonal sums are excluded from weak Sidonicity. The capacity lemma therefore gives

$$k^2 \leq (N/T + 2/3 + 200e^{-\alpha N/T})(T + 4k).$$

Put $x = N^{1/4} \geq 90$, $T = \sqrt{6}x^3$ and $\beta = 2\sqrt{6}/3 = \sqrt{8/3}$. Then $T \leq N$ and, writing $\varepsilon = 200e^{-\alpha x/\sqrt{6}}$,

$$P_\varepsilon(k) \leq 0, \quad P_\varepsilon(z) = z^2 - (\beta x + \beta^2 + 4\varepsilon)z - x^4 - \beta x^3 - \sqrt{6}\varepsilon x^3. \quad (4.3)$$

The derivative of $\log u - 2(u - 1)/(u + 1)$ is $(u - 1)^2/[u(u + 1)^2] \geq 0$, so $\alpha \geq 2/7$. As $\sqrt{6} < 49/20$, $\alpha/\sqrt{6} \geq 40/343$ and monotonicity gives $\varepsilon x \leq 18000e^{-3600/343} < 1/2$. The strict last comparison has the exact rational certificate

$$(1957/720)^{10} \sum_{j=0}^4 \frac{(170/343)^j}{j!} > 36000, \quad 3600/343 = 10 + 170/343.$$

Both factors are lower bounds for the corresponding exponentials by positive series. Clearing denominators gives the integer comparison

$$\begin{aligned} & 449233538381596942864443106714320805345633416 \\ & > 447728960659239231243811144335360000000000000. \end{aligned}$$

For $y = x^2 + \beta x + 2$,

$$P_0(y) = \frac{4}{3}x^2 - \frac{2}{3}\beta x - \frac{4}{3}, \quad \frac{P_0(y)}{x^2} \geq \frac{8024}{6075} > \frac{13}{10},$$

where $\beta < 5/3$ and $x \geq 90$ were used. Also

$$\frac{\varepsilon(4y + \sqrt{6}x^3)}{x^2} < \frac{49}{40} + \frac{2}{x} + \frac{10}{3x^2} + \frac{4}{x^3} \leq \frac{49}{40} + \frac{1}{45} + \frac{1}{2430} + \frac{1}{182250} < \frac{5}{4}.$$

Thus $P_\varepsilon(y) > x^2/20 > 0$. The positive-root argument from Lemma 4.1 proves Theorem 1.2.

4.2 Bounded difference multiplicity

For the hypothesis of Theorem 1.3, ordered-pair expansion and the lattice majorant give, at every $0 < T \leq N$,

$$E_f(\mu, \mu) = \frac{4}{3}k + 2 \sum_{d \geq 1} r_A(d) f(d/T) \leq \frac{4}{3}k + gT.$$

Only the nonzero differences receive the factor g . Substitute $S = gT$, $X = gN$ in the capacity inequality; every $0 < S \leq X$ is allowed. It becomes (4.1), and Lemma 4.1 proves (1.4). Translation to the closed interval $[0, D]$ proves the diameter version. Empty and singleton sets are immediate; that version requires $D > 0$. For example, if $k \geq 20366$, counting gives $gD \geq k(k-1)/2 \geq 207376795 > 120^4$, so the explicit inverse is

$$gD > \left(\frac{\sqrt{4(k-1) + 8/9} - 2\sqrt{2/3}}{2} \right)^4.$$

4.3 Difference triangle sets

Let the scope be m and put $K = k + 1$. For $0 < T \leq m$, apply the capacity lemma separately to each row measure $\mu_i = \sum_{j=0}^k \delta_{a_{ij}/T}$. Summation, using the single shared budget for all positive within-row differences, gives

$$nK^2 \leq C(m/T) \sum_i E_f(\mu_i, \mu_i) \leq C(m/T)(4nK/3 + T).$$

With $S = T/n$, $X = m/n$, division by n gives (4.1) for K , for every real $0 < S \leq X$. No cross-row differences are inserted and no integrality of X is needed. Counting the $n \binom{k+1}{2}$ different positive integers yields

$$X \geq k(k+1)/2, \quad 20364 \cdot 20365/2 = 207356430 < 120^4 < 207376795 = 20365 \cdot 20366/2.$$

Thus $k \geq 20365$ allows Lemma 4.1. It gives $k < X^{1/2} + \gamma X^{1/4}$, with $\gamma = 2\sqrt{2/3}$. Inverting the increasing quadratic proves $m > n\Phi(k)^4$. For an explicit expansion put $z = \Phi(k)$, so $z^2 + \gamma z = k$. Then

$$z^4 = k^2 + \gamma^2 k - \gamma(2k + \gamma^2)z, \quad z \leq \sqrt{k} - \gamma/2 + \frac{\gamma^2}{8\sqrt{k}}.$$

The latter follows from $\sqrt{4k + \gamma^2} \leq 2\sqrt{k} + \gamma^2/(4\sqrt{k})$. Substitution gives

$$z^4 \geq k^2 - 2\gamma k^{3/2} + 2\gamma^2 k - \frac{5\gamma^3}{4}\sqrt{k} + \frac{\gamma^4}{2} - \frac{\gamma^5}{8\sqrt{k}} \geq k^2 - \frac{4\sqrt{2}}{3}k^{3/2} + \frac{16}{9}k - 2\sqrt{k}.$$

For the last step use $\gamma < 1$, $k \geq 1$, and $5\gamma^3/4 + \gamma^5/(8k) < 2$, discarding the positive constant. This proves both statements in Theorem 1.4.

5 Product kernels

Let ρ be the pushforward of $\nu_{R/T}$ under $s \mapsto Ts$, where $0 < T \leq R$. For $F_T(z) = \prod_{j=1}^d f(z_j/T)$, the signed product measure $\rho^{\otimes d}$ has potential one on $[0, R]^d$ and energy at most $(R/T + b + 200e^{-\alpha R/T})^d$. This follows by Fubini; each factor energy is nonnegative. The kernel is the autocorrelation of $T^{-d/2} \prod_j h(z_j/T)$, so signed Cauchy-Schwarz is valid. A positive point measure μ of mass k in the box therefore obeys

$$k^2 \leq (R/T + b + 200e^{-\alpha R/T})^d E_{F_T}(\mu, \mu). \quad (5.1)$$

The certificate need not be supported in the box; its exact potential on the box and its finite total variation are what is used.

5.1 Manhattan diameter and the index-two lattice

The map $(x, y) \mapsto (u, v) = (x + y, x - y)$ satisfies $|\Delta x| + |\Delta y| = \max(|\Delta u|, |\Delta v|)$. For a nonempty configuration translate each new coordinate separately to lie in $[0, r]$. The transformed points may lie in either coset of $\Lambda = \{(u, v) \in \mathbb{Z}^2 : u \equiv v \pmod{2}\}$, but every difference lies in Λ and remains unique. Empty configurations are immediate. Put $a = 4/3$ and

$$S_e = \sum_{j \in \mathbb{Z}} f(2j/T), \quad S_o = \sum_{j \in \mathbb{Z}} f((2j+1)/T).$$

Monotonicity gives $S_e \leq T/2 + a$. With $v_0 = f(1/T)$, comparison with the preceding intervals gives $S_o \leq 2v_0 + T \int_{1/T}^{\infty} f \leq T/2 + v_0 \leq T/2 + a$; this is valid also for $T < 1$ when the relevant samples vanish. The complete lattice sum is $S_e^2 + S_o^2$. Its zero-vector term is a^2 , while the point measure has m such diagonal terms. Thus

$$m^2 \leq \left(r/T + b + 200e^{-\alpha r/T}\right)^2 (a^2 m + T^2/2 + 2aT + a^2), \quad 0 < T \leq r. \quad (5.2)$$

This explicitly retains the lower-order lattice terms.

For $A \geq 0$, $B > 0$, the positive root of $z^2 - Az - B$ is at most

$$\sqrt{B} + A/2 + A^2/(8\sqrt{B}). \quad (5.3)$$

It follows from $\sqrt{B + A^2/4} \leq \sqrt{B} + A^2/(8\sqrt{B})$. Put $x = r^{1/3}$, $c = (4/3)^{1/3}$, $t = \sqrt{2}c$, $T = tx^2$. Then $1 < t < 2$, $c < 4/3$. For $x \geq 160$, the scale is valid and

$$\varepsilon x = 200xe^{-\alpha x/t} \leq 200xe^{-x/8} \leq 32000e^{-20} < 32000/2^{20} < 1/32.$$

Write $\beta = b + \varepsilon < 1$, $D = x/t + \beta \leq 2x$ and $B_0 = T^2/2 + 2aT + a^2$. Since $T/\sqrt{2} \leq \sqrt{B_0} \leq T/\sqrt{2} + \sqrt{2}a$, (5.3) applied to (5.2) yields

$$m \leq D(T/\sqrt{2} + \sqrt{2}a) + a^2 D^2/2 + a^4 \sqrt{2} D^3/(8T).$$

The leading terms are $x^3/\sqrt{2} + cx^2$, since $bt/\sqrt{2} + a^2/(2t^2) = c$ (the summands are $2c/3$ and $c/3$). The remaining terms obey

$$\begin{aligned} \varepsilon tx^2/\sqrt{2} &\leq x/24, \\ a^2 \beta x/t &\leq (16/9)x, \\ \sqrt{2}ax/t &\leq 2x, \\ a^2 \beta^2/2 + \sqrt{2}a\beta &\leq 26/9 \leq (13/720)x, \\ a^4 \sqrt{2} D^3/(8T) &\leq (128/27)x. \end{aligned}$$

Their sum is at most $(18529/2160)x < 9x$, proving Theorem 1.5. In particular an unspecified lower-order error has not been silently deleted.

5.2 Integer boxes and a sharper lattice estimate

For $T \geq 1$ let $n_0 = \lfloor T \rfloor$, $\eta = T - n_0$ and $S(T) = \sum_{j \in \mathbb{Z}} f(j/T)$. Finite sums of first and third powers give

$$\begin{aligned} S(T) &= \frac{4}{3} + \frac{8}{3}n_0 - \frac{2n_0(n_0+1)}{T} + \frac{n_0^2(n_0+1)^2}{3T^3} \\ &= T + \frac{1}{3T} - \frac{2\eta(1-\eta)(1-2\eta)}{3T^2} + \frac{\eta^2(1-\eta)^2}{3T^3} \leq T + \frac{7}{16T} \leq T + \frac{1}{2T}. \end{aligned} \quad (5.4)$$

Indeed $\eta(1-\eta) \leq 1/4$ and $\eta(1-\eta)|1-2\eta| \leq 1/8$: with $u = |1-2\eta|$, the latter is $u(1-u^2)/4 \leq u(1-u)/2 \leq 1/8$. An integral endpoint contributes zero, so no separate rounding case is omitted.

Unique nonzero ordered vector differences in $[N]^d$ and (5.1) imply

$$k^2 \leq D^d[a^d k + S(T)^d - a^d] \leq a^d D^d k + D^d(T + 1/(2T))^d, \quad D = N/T + b + 200e^{-\alpha N/T}, \quad (5.5)$$

for $1 \leq T \leq N$. The diagonal is $a^d k$, and the full lattice budget includes the origin exactly once, explaining the subtraction. No factor 2^d is added to an already ordered difference count.

Use the x, c_d of (1.8), and put

$$t = (a^d/b)^{1/(d+1)}, \quad s = bt = (ab)^{d/(d+1)}, \quad q = x^{-1}, \quad K = N^{d/2}, \quad T = tN/x.$$

Then $1 < t < 3/2$, $0 < s < 1$, $a^d/t^d = s$ and $T \geq x$. Under $x \geq \max(120, 4d)$ we have $T \leq N$, $q \leq 1/120$, $dq \leq 1/4$. As before $\varepsilon = 200e^{-\alpha x/t}$ satisfies $\varepsilon x < 24000/2^{20} < 1/32$. Define

$$w = (b + \varepsilon)tq, \quad z = 1/(2T^2), \quad v = (1 + w)(1 + z) - 1.$$

Then

$$0 \leq w \leq sq + 3q^2/64 \leq q + q^2, \quad 0 \leq z \leq q^2/2, \quad 0 \leq v \leq sq + q^2 \leq q + q^2.$$

For the last inequality the excess coefficient is at most $3/64 + 1/2 + (q + q^2)/2 \leq 45/64 < 1$, already for $q \leq 1/4$. Hence $dw, dv \leq 5/16 < 1/3$. The binomial estimate $(1 + u)^d \leq (1 - du)^{-1}$, $du < 1$, gives

$$(1 + w)^d \leq 3/2, \quad (1 + v)^d \leq 3/2, \quad (1 + w)^d - 1 \leq dw/(1 - dw) \leq 2dq.$$

Dividing (5.5) by K^2 gives a quadratic for $U = k/K$, with linear coefficient $sq(1 + w)^d$ and constant term $(1 + v)^d$; here $x^{d+1} = K$. Its root bound is

$$U \leq (1 + v)^{d/2} + \frac{sq}{2}(1 + w)^d + \frac{s^2q^2(1 + w)^{2d}}{8(1 + v)^{d/2}}. \quad (5.6)$$

The final term is at most $9q^2/32$ and the middle term at most $sq/2 + dq^2$. Taylor's theorem, with $p = d/2 \geq 1$, bounds the second derivative of $(1 + v)^p$ by $(3/2)p(p - 1)$ on the relevant interval; for $1 \leq p < 2$ use $(1 + u)^{p-2} \leq 1$. Since $v \leq sq + q^2 \leq 2q$,

$$(1 + v)^{d/2} \leq 1 + (d/2)sq + (d/2 + 3d^2/4)q^2.$$

Consequently

$$U \leq 1 + \frac{d+1}{2}sq + \left(\frac{3d^2}{4} + \frac{3d}{2} + \frac{9}{32}\right)q^2 \leq 1 + c_dq + 2d^2q^2.$$

The final comparison holds for all $d \geq 2$ because $5d^2/4 - 3d/2 - 9/32$ is positive at 2 and increasing. Multiplication by K proves (1.9). AM-GM on $(dbt + a^d t^{-d})/2$ explains the chosen scale and coefficient. This optimizes the retained product inequality only.

5.3 A two-dimensional bound for every positive integer

For $A \subseteq [N]^2$, take an integer $T \geq 1$ and $r(z) = \#\{a \in A : z \in a + [T]^2\}$. It is supported on $[N + T - 1]^2$. Write $M = N + T - 1$. Window intersections give exactly

$$\sum_z r(z) = kT^2, \quad \sum_z r(z)^2 = kT^2 + \sum_{a \neq b} (T - |a_1 - b_1|)_+(T - |a_2 - b_2|)_+.$$

Since $\sum_{j \in \mathbb{Z}} (T - |j|)_+ = T^2$, uniqueness gives the full nonzero-vector budget $T^4 - T^2$. Thus

$$\sum_z r(z)^2 \leq T^4 + (k - 1)T^2, \quad k^2 \leq M^2[1 + (k - 1)/T^2]. \quad (5.7)$$

The last step is finite Cauchy-Schwarz on M^2 positions. Empty sets are covered as well, because $T^4 - T^2 \geq 0$. Put $x = N^{1/3} \geq 1$, $T = \lceil x^2 \rceil$ and $A_0 = M^2/T^2$. Then $x^3 \leq M \leq x^3 + x^2$, $A_0 \leq (x + 1)^2$. Drop the negative constant $-A_0$ in the quadratic and use (5.3) to get

$$\begin{aligned} k &\leq M + A_0/2 + A_0^2/(8M) \\ &\leq x^3 + \frac{3}{2}x^2 + \frac{9}{8}x + 1 + \frac{3}{4x} + \frac{1}{2x^2} + \frac{1}{8x^3} \leq x^3 + \frac{3}{2}x^2 + \frac{7}{2}x. \end{aligned}$$

The final difference, multiplied by $8x^3$, is

$$19x^4 - 8x^3 - 6x^2 - 4x - 1 = (x - 1)(19x^3 + 11x^2 + 5x + 1) \geq 0.$$

Let $c = (8/3)^{1/3}$. At $x \geq 120$, (1.9) with $d = 2$ gives $k \leq x^3 + cx^2 + 8x$. For $1 \leq x < 120$, the integer comparison $83^3 = 571787 < 576000 = (8/3)60^3$ gives $c > 83/60$, hence $(3/2 - c)x^2 < (7/60)x^2 < 14x$. The elementary bound is therefore less than $x^3 + cx^2 + (35/2)x \leq x^3 + cx^2 + 18x$. This proves (1.10), including $N = 1$ and the splice point $N = 120^3$.

6 Verification status and limits

The transfer scout derivations were supplied by Claude (Anthropic). GPT-6 Astra (OpenAI) independently examined those derivations, repaired the combinatorial and remainder arguments where needed, and wrote the complete explicit proofs. The cosine functional proof and admissible perturbation were derived in this informed proof-writing process, followed by independent OpenAI in-team review. These transfers were not clean-room attacks. The author takes responsibility for the mathematics.

Three isolated Claude reports have now been received. The cosine sonar theorem, weak Sidon, bounded multiplicity, difference triangles, triangular sonar, Manhattan and general box bounds, and the kernel functional and perturbation have passed their respective mathematical reviews. The perturbed sonar consequence passed with editorial repairs, incorporated here. The all- N square supplement has OpenAI in-team review and source-formula checking, but was not in the Claude referees’ reading lists.

The referee verdicts are mathematical model reviews, not independent human referee reports. Source-formula checking of this manuscript is separate from those reviews. No new Lean formalisation is claimed for sonar, weak Sidon, bounded multiplicity, difference triangles, Manhattan configurations, boxes, or the kernel functional. The earlier ordinary Sidon theorem

$$|A| \leq \sqrt{N} + (2\sqrt{2}/3)N^{1/4} + 1, \quad N \geq 120^4,$$

with diagonal sums included, has been kernel-checked in Lean v4.34.0-rc1 with Mathlib commit `de5ce8a9`. The public source at the checked revision is available in the accompanying repository. The final theorem’s reported axioms are only `propext`, `Classical.choice`, and `Quot.sound`. This formalisation concerns that specific finite Sidon inequality; it does not automatically formalise any new analytic transfer here.

The dependency-free JavaScript checkers supplement the proofs with exact BigInt arithmetic. They check finite combinatorial conventions, lattice identities, scale substitutions, and rational error envelopes. The recorded executions all returned PASS:

Checker subject	Executed finite scope
Sonar	820 marginal identities, 1070 sonar prefixes, 41856 energy cases; non-sonar negative control rejected.
Weak Sidon	16384 subsets, 2048 weak Sidon sets, 16384 energy cases; 1558 repeated-difference instances.
Bounded multiplicity	4096 subsets, 7521 set/parameter pairs, 60168 energy cases and 54 substitutions.
Difference triangles	687 row families, 4122 energy cases, 72 substitutions including 54 nonintegral scope quotients.
Manhattan	4096 subsets, 721 configurations, 3605 energy cases, 72 parity-lattice cases.
Boxes	768 subsets in dimensions two and three, 341 Sidon sets, 1705 energy cases; symbolic lattice identity and 292 rational scales.
All- N square supplement	530 subsets, 3180 window identities, 1194 Sidon energy bounds, 1004 ceiling checks and two polynomial identities.
Cosine and perturbation	16 and 29 exact rational checks, respectively, including coefficient, Fourier-positivity, tail and denominator margins.

The six principal combinatorial checkers cover 127840 energy-accounting cases. These counts are not evidence for the capacity inequality itself: the Manhattan and box checkers do not evaluate its certificate, and the sonar checks at small parameters use a tail term so large that they cannot meaningfully stress the capacity step. Their useful scope is the exact marginal identities, lattice sums, combinatorial accounting and final scalar algebra. The mathematical capacity proof is given in Section 2. The referees additionally report numerical certificate and finite-capacity tests; those are floating-point evidence, not proofs, and were not independently rerun for this manuscript. These finite checks do not prove an infinite-family statement or replace the analytic arguments. The cosine checker does not perform symbolic integration; the integral identities are proved in the text. The perturbation checker checks the rational envelopes in its analytic admissibility proof, not an assertion of optimality from a grid.

The literature comparisons use a bounded source audit dated October 2, 2026. Its primary and secondary readings are distinguished in Section 1; unread relevant sources are named there. Absence of an earlier matching statement in that audit is not a priority claim. No optimality is asserted for the displayed onsets or remainder constants. The full horizontal functional infimum remains open, and the product-template optimization does not exclude different two-dimensional kernels or additional combinatorial information.

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